

MINISTRY OF AYUSH • NATIONAL MEDICINAL PLANTS BOARD (NMPB)

CENTRAL SECTOR SCHEME FOR SCHOOL HERBAL GARDENS

SCHOOL AUSHADHI LAB

Decentralized Living Herbal Sanctuaries, Ethnobotanical Bio-Commons,
and Planetary One Health Pedagogy across 50 Schools of Manipur

*A Modern Strategic Transformation of the Baseline NMPB School Herbal Garden Initiative (2026–2030)
Designed under the Ethos of Nordic Lagom Functional Minimalism & Zero-Waste Circularity*

Implementing Agency:	ONE VISION MANIPUR (Regd. No. 1 of 1988 M Darpan ID: MN/2024/0394690)
Institutional Seat:	Sagolband Tera Sapam Leirak, Nongmaithem Leikai, Imphal, Manipur - 795001
President:	Oinam Thoiba Singh (President, One Vision Manipur)
Project Leadership:	Sapam Chital Singh (PI), Lamnganbi Ngangom (Coordinator), Thaibal Singh (General Secretary)
Target Reach:	50 Verified Educational Campuses across Imphal East, Imphal West & Valley Corridors
Physical Footprint:	25,000 m² Aggregate Dedicated Land (500 m² Protected Enclosure per Campus)
Project Period & Budget:	5 Years (2026–2030) Total Statutory NMPB Grant: ₹26,50,000
Global Alignment:	WHO Global Traditional Medicine Strategy 2025–2034 • One Health Framework

Imphal, Manipur • September 2026

1. Executive Summary & The Lagom Framework

School Aushadhi Lab fundamentally shifts school herbal gardens from passive horticulture into dynamic, decentralized **Living Ethnobotanical Sanctuaries**. Positioned within the ecologically sensitive Indo-Burma Biodiversity Hotspot, this project coordinates 50 selected schools across the Imphal Valley to establish 25,000 square meters of high-density medicinal micro-reserves. The initiative is executed by **One Vision Manipur**, an established non-profit founded in 1988 under the presidency of **Oinam Thoiba Singh**, in direct partnership with the National Medicinal Plants Board (NMPB), Ministry of AYUSH.

The project integrates two authoritative paradigms: the community-driven ecological defense ethos of **Greenpeace** and the systemic, evidence-based public health rigor of the **World Health Organization (WHO)**. Grounded in the *WHO Global Traditional Medicine Strategy 2025–2034*, it bridges ancient Meitei *Hidaklon* healing wisdom with modern botanical science and climate-resilient permaculture.

THE NORDIC LAGOM DESIGN PRINCIPLE (*Inte för mycket, inte för lite, utan precis lagom*)

- **Functional Equilibrium:** Every plant, structure, and pathway fulfills a specific ecological, educational, or therapeutic purpose. Unnecessary ornamental clutter and synthetic ornamentation are strictly eliminated.
- **Organic Materiality:** Construction utilizes local seasoned bamboo, non-hewn river stones, permeable gravel, and natural pine resin treatments, harmonizing built elements with local soil biology.
- **Circular Metabolism:** Zero-waste resource flows cycle on-site biomass through three-chamber vermicomposting, bio-char soil remediation, and rainwater recharge bioswales (15,000-liter seasonal retention per school).
- **Spacious Typographic & Visual Restraint:** Clear spatial zoning gives each specimen room to thrive while providing uncrowded, calm pedagogical spaces for open-air inquiry.

Core Strategic Goals

- 1. In-Situ Conservation:** Establish permanent gene banks protecting 20 critical medicinal taxa, with special emphasis on vulnerable endemic species like *Paris polyphylla* and *Curcuma angustifolia*.
- 2. Pedagogical Transformation:** Train 5,000 certified 'One Health Junior Fellows' across 50 secondary schools, integrating hands-on ethnobotany into the National Education Policy (NEP) curriculum.
- 3. Nutritional Security:** Synergize garden yields (Moringa, Amla, Tikhur) directly with the PM POSHAN Mid-Day Meal Scheme to address pediatric anemia and micronutrient deficiencies.
- 4. Biocultural Safeguarding:** Connect youth with traditional Meitei healers (*Maibas* and *Maibis*) to preserve oral indigenous pharmacology through interactive digital QR field tagging.

2. Spatial Architecture & Biophysical Layout (500 m²)

Each 500 m² campus plot is partitioned into five specialized ecological zones surrounding a central timber agora. This geometry provides clear sightlines, efficient micro-irrigation, and zoned microclimates tailored to the physiological demands of diverse herbal species.

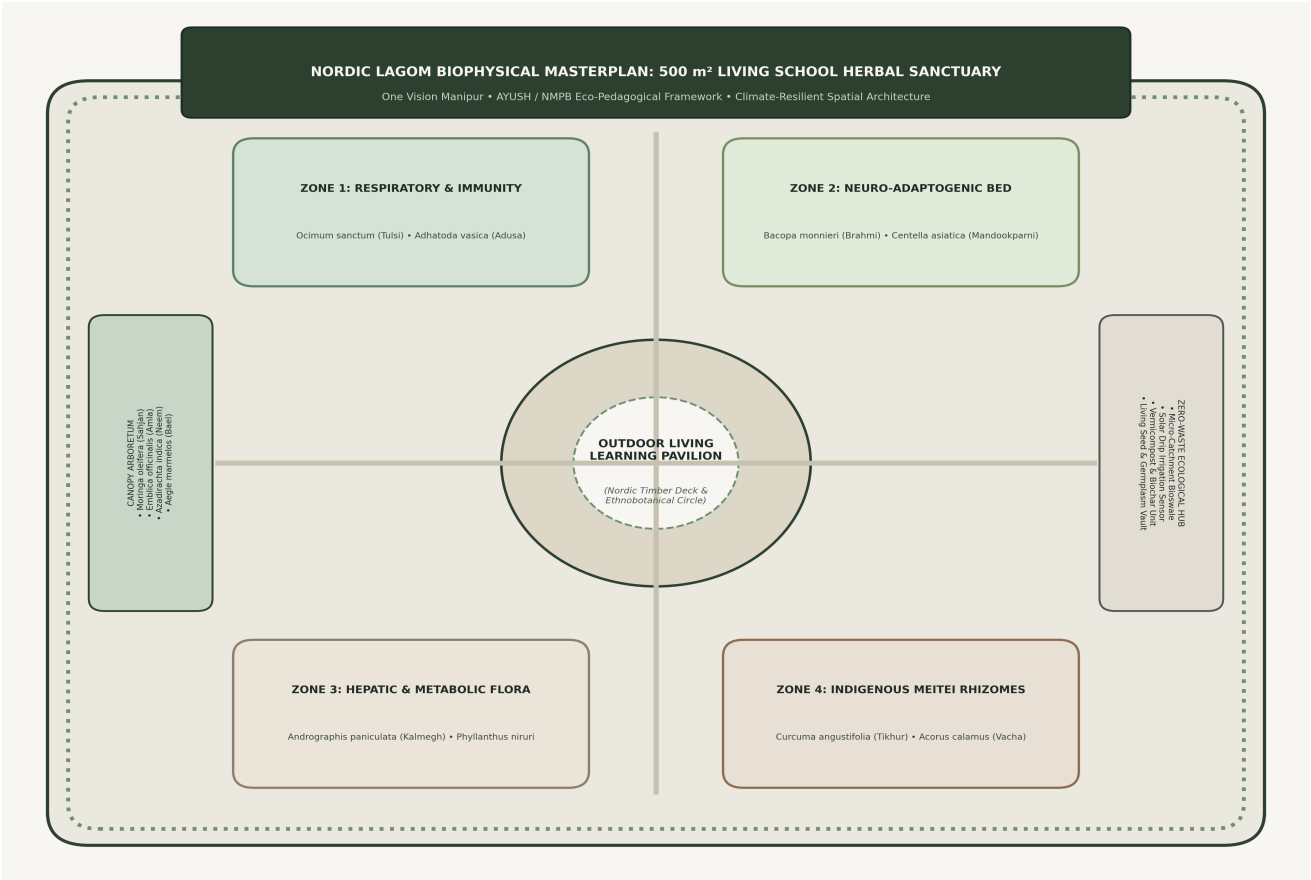


Figure 1: Nordic Lagom Biophysical Masterplan (25m x 20m) showing zonal beds, central timber learning pavilion, permeable pathways, and ecological support units.

Zonal Functional Breakdown

Zone / Feature	Dimensions & Microclimate	Key Cultivated Taxa	Pedagogical & Clinical Purpose
Central Timber Agora	Radius 3.0m (28 m²) Raised timber deck	Native climbing vines (*Shatavari*)	Open-air Socratic amphitheater for 35 students; ethnobotanical seminars.
Zone 1: Respiratory & Immune	6.5m x 3.2m (21 m²) Sunny, well-drained loam	*Ocimum sanctum* (Tulsi) *Adhatoda vasica* (Adusa)	Anti-tussive and bronchodilatory study; school infirmary herbal teas.
Zone 2: Neuro-Adaptogenic	6.5m x 3.2m (21 m²) Moist, semi-shaded basin	*Bacopa monnieri* (Brahmi) *Centella asiatica* (Peruk)	Cognitive tonic education; moisture-retentive wetland soil ecology.
Zone 3: Hepatic & Metabolic	6.5m x 3.2m (21 m²) Friable sandy loam	*Andrographis paniculata* *Phyllanthus niruri*	Bitter alkaloid study; hepatic health and seasonal viral prophylaxis.
Zone 4: Meitei Rhizomes	6.5m x 3.2m (21 m²) Deep rich leaf mould	*Curcuma angustifolia* (Tikhur) *Acorus calamus* (Vacha)	Geophyte reproduction; traditional starch processing and gut health.
Perimeter Arboretum	North & West borders Deep soil root barrier	*Moringa oleifera*, *Amla*, *Neem*, *Aegle marmelos*	Canopy windbreak, particulate capture, and continuous midday meal nutrition.

3. Ethnobotanical & Pharmacological Taxa Matrix

The project deploys 20 scientifically validated medicinal taxa that combine therapeutic relevance, educational value, and resilience within Manipur's climate. The selection honors indigenous Meitei medical nomenclature (*Hidak*) alongside Ayurvedic and modern pharmacological classifications.

Botanical Name	Meitei Name	Active Bioactives	Clinical & Nutritional Indication
<i>Ocimum sanctum</i>	Tulsi	Eugenol, Ursolic acid	Antimicrobial, adaptogenic respiratory defense.
<i>Adhatoda vasica</i>	Nongmangkha	Vasicine, Vasicinone	Potent bronchodilator and expectorant.
<i>Andrographis paniculata</i>	Kalmegh	Andrographolide	Hepatoprotective, immune stimulant, antipyretic.
<i>Phyllanthus niruri</i>	Chakpa-heikru	Phyllanthin, Nirurin	Hepatic clearance, antiviral, urinary tract support.
<i>Centella asiatica</i>	Peruk	Asiaticoside, Madecassoside	Neuro-regenerative, wound healing, anti-inflammatory.
<i>Bacopa monnieri</i>	Brahmi	Bacosides A & B	Synaptic facilitator, memory and cognitive enhancement.
<i>Curcuma angustifolia</i>	Yaingang-manbi	Curcuminoids, Starch complex	Demulcent, gastrointestinal restorative, weaning tonic.
<i>Acorus calamus</i>	O-khihidak	alpha- & beta-Asarone	Nootropic carminative; speech and vocal clarity.
<i>Moringa oleifera</i>	Chigonglei-machu	Moringine, Quercetin	Superfood foliage for PM POSHAN micronutrient enrichment.
<i>Phyllanthus emblica</i>	Heikru	Ascorbic acid, Emblicanin	Antioxidant cellular protector; natural Vitamin C source.
<i>Azadirachta indica</i>	Neem	Azadirachtin, Nimbin	Antiseptic, dermatological wash, organic pesticide.
<i>Aegle marmelos</i>	Heirikhagok	Marmelosin, Luvangetin	Chronic bowel regularity and gastrointestinal balance.
<i>Gymnema sylvestre</i>	Sing-nang-manbi	Gymnemic acids	Inhibits intestinal glucose uptake; metabolic support.
<i>Hemidesmus indicus</i>	Anantmool	Hemidesmin, Aldehydes	Systemic blood cleanser and cooling anti-inflammatory.
<i>Oroxylum indicum</i>	Shayon	Baicalein, Chrysin	Anti-inflammatory Dashmoola constituent; sacred grove tree.
<i>Aloe barbadensis</i>	Ghritkumari	Acemannan, Barbaloin	Epithelial soothing, hydration, emergency skin burn relief.
<i>Asparagus racemosus</i>	Mayang-chigong	Shatavarins I–IV	General vitality, adaptogenic endocrine support.
<i>Catharanthus roseus</i>	Sadabahar	Vinblastine, Vincristine	Alkaloid pharmacology study; anti-diabetic use.
<i>Litsea cubeba</i>	Mei-ngang	Citral, Limonene	Antifungal essential oils; traditional aromatherapy.
<i>Paris polyphylla</i>	Sing-khei	Polyphyllin D, Dioscin	Vulnerable endemic Himalayan root; in-situ germplasm banking.

4. Pedagogical Integration & One Health Impact

School Aushadhi Lab translates the *WHO One Health Joint Plan of Action* into an actionable school curriculum. The health of students, the vitality of soil microbial communities, and the protection of native pollinators are treated as interconnected elements of a single living ecosystem.

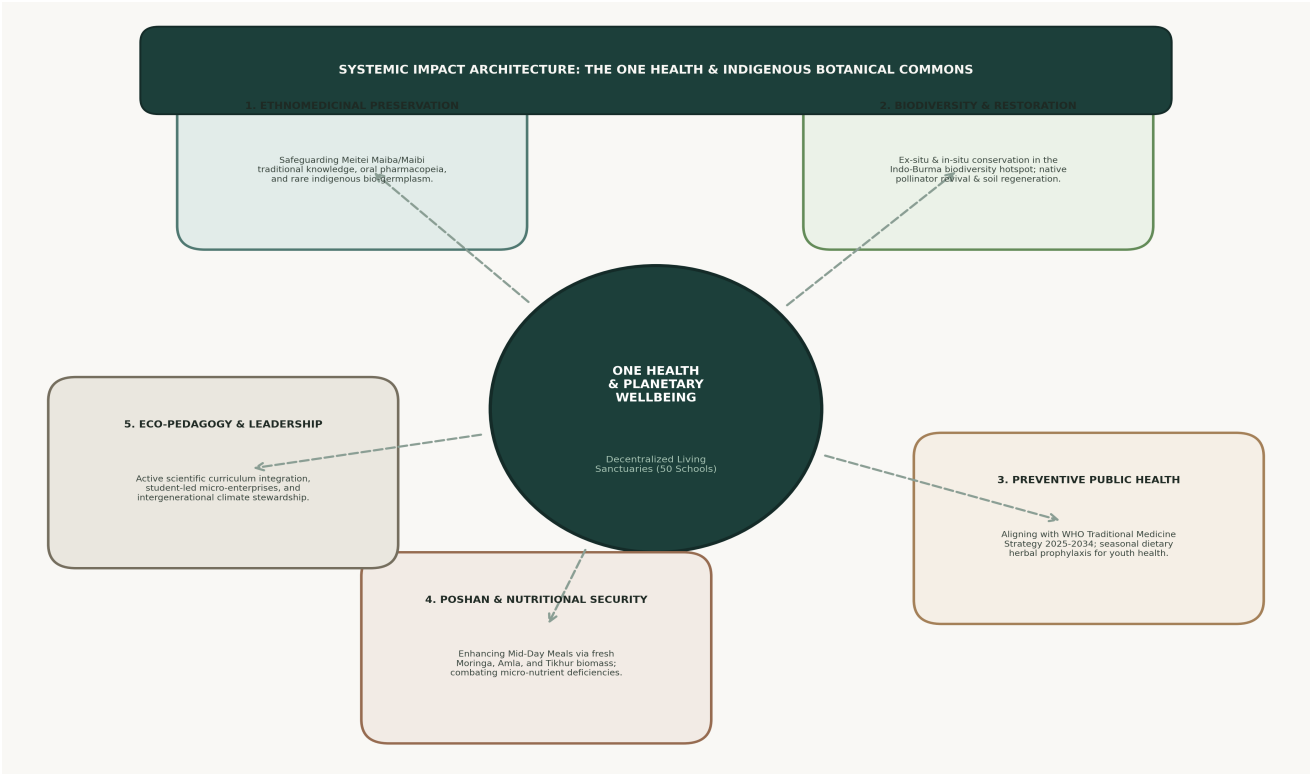


Figure 2: Systemic Impact Architecture linking ethnomedicinal preservation, biodiversity revival, preventive health, PM POSHAN nutritional security, and youth eco-pedagogy.

Pedagogical Enablers & Learning Protocols

- **The 'One Health Junior Fellows' Leadership Cohort:** A selective group of 20 students per school (1,000 students across the network) receives structured training in soil chemistry, phenological data logging, botanical taxonomy, and natural bio-fertilizer formulation.
- **Weather-Resistant QR Field Markers:** Every bed features bamboo laser-engraved QR markers linking to an open-access bilingual digital repository. Scanning a marker reveals chemical structures, pharmacological effects, folk medicine traditions, and home cultivation guides.
- **PM POSHAN Nutrition Enrichment:** Daily school lunches are systematically supplemented with freshly harvested Moringa foliage and Amla, supplying vital iron, calcium, and vitamin C to improve student health and attendance.

5. Governance & 5-Year Implementation Roadmap

To prevent garden abandonment during holidays and guarantee long-term survival, governance is anchored by the **Herbal Garden Management Committee (HGMC)** at each school. Comprising the Principal, Eco-Club Coordinator, Village Authority, PTA, two student leads, and an elder Meitei *Maiba/Maibi*, the HGMC provides community oversight and continuous care.

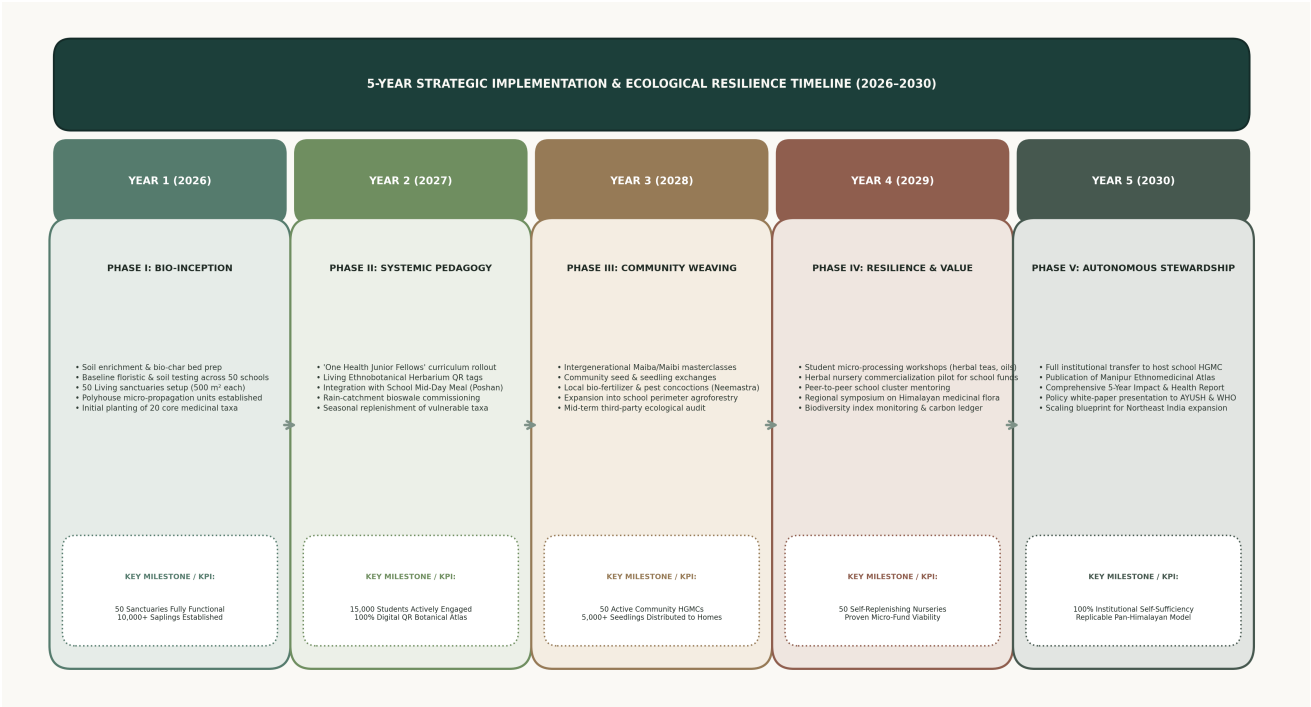


Figure 3: Five-Year Phased Implementation & Resilience Roadmap (2026–2030) from initial bio-inception to full autonomous school stewardship.

Phased Timeline Summary

Phase & Year	Focus Area	Key Deliverables & Operations	Primary KPI
Year 1 (2026) Bio-Inception	Ground establishment & planting	Soil remediation with bio-char, bamboo fencing, bioswale digging, planting 10,000 hardened saplings.	50 Functional Sanctuaries; > 85% sapling survival.
Year 2 (2027) Systemic Pedagogy	Curriculum rollout & nutrition	Installing QR tags, launching Junior Fellows program, integrating Moringa into PM POSHAN meals.	1,000 Fellows trained; 100% QR tagging active.
Year 3 (2028) Community Weaving	Elder integration & bio-pesticides	Hosting 200 Maiba/Maibi masterclasses, running seed exchanges, spraying organic <i>Neemastra</i> .	200 Elder workshops; 5,000 seedlings distributed.
Year 4 (2029) Value Creation	Herbal processing & micro-funds	Training students in solar leaf drying and herbal teas; launching nursery seedling sales for school funds.	50 Self-funding nurseries; Peer mentoring launched.
Year 5 (2030) Autonomy	Institutional handover & scale	Publishing the Manipur Ethnobotanical Atlas, final financial audits, transferring 100% management to HGMC.	100% School ownership; Statewide policy brief.

6. Financial Architecture & Statutory Grant Breakdown

The project strictly complies with the financial norms of the **National Medicinal Plants Board (NMPB)** under the Central Sector Scheme for School Herbal Gardens. Grant funds are augmented by an innovative circular micro-financing mechanism that ensures lasting sustainability beyond the five-year grant period.

Consolidated Statutory 5-Year Budget (50 Schools)

Fiscal Year	Funding Category	Per School Grant (₹)	Total for 50 Schools (₹)	Statutory Source
Year 1 (2026)	Establishment, Land Prep & Planting	₹25,000	₹1,250,000	NMPB Central Grant
Year 2 (2027)	Annual Maintenance & Replenishment	₹7,000	₹350,000	NMPB Central Grant
Year 3 (2028)	Annual Maintenance & Replenishment	₹7,000	₹350,000	NMPB Central Grant
Year 4 (2029)	Annual Maintenance & Replenishment	₹7,000	₹350,000	NMPB Central Grant
Year 5 (2030)	Annual Maintenance & Transition	₹7,000	₹350,000	NMPB Central Grant
Total Funding	Consolidated 5-Year Budget Request	₹53,000	₹2,650,000	Ministry of AYUSH

Itemized Year 1 Setup Breakdown (Per 500 m² School Plot = ₹25,000)

Component Category	Technical Operational Scope	Allocated (₹)	Share (%)
Soil Remediation & Biology	Deep trenching, 3 tonnes organic compost, bio-char inoculation, and earthworm release.	₹5,500	22.0%
Certified Saplings & Seedlings	Procurement of 200 certified, disease-free nursery saplings and rootstocks across 20 taxa.	₹7,000	28.0%
Eco-Paths & Bioswale	Untreated bamboo bed edging, permeable gravel paths, and excavation of 15k-L recharge bioswale.	₹4,500	18.0%
Nordic Timber Agora	Raised central bamboo seating ring and wooden observation platform (open-air learning center).	₹3,500	14.0%
Pedagogical Equipment	Weatherproof bamboo QR codes, botanical magnifying lenses, soil testing reagents, and seed jars.	₹2,500	10.0%
Orientation & Elder Honoraria	Faculty orientation seminar, Maiba honorarium for garden blessings, and Junior Fellow workbooks.	₹2,000	8.0%
Total Per Campus	Full capital requirement for complete Year 1 setup	₹25,000	100.0%

Annual Maintenance Breakdown (Years 2–5: ₹7,000 / Year)

- **Biological Replenishment (₹2,500):** Replacing expired plants, root division, and propagating heirloom seed varieties.
- **Organic Inputs & Nutrition (₹2,000):** Replenishing vermicompost beds, microbial *Jeevamrutha* inoculants, and bio-fertilizer cakes.
- **Tool Upkeep & Water Lines (₹1,500):** Maintaining drip lines, servicing nozzles, replacing shears, and clearing bioswale channels.
- **Community Workshops & Record Books (₹1,000):** Supplying field journals, photographic logs, and honoraria for visiting healers.

7. Impact Verification & Institutional Mandate

To ensure complete transparency and measurable outcomes, School Aushadhi Lab deploys a dual monitoring system combining digital tracking with regular third-party field evaluations.

Target Metric	Baseline (Day 0)	Year 1 Target	Year 3 Target	Year 5 Final Target
Active Sanctuaries	0	50 Gardens	50 Gardens	50 Fully Transferred
Healthy Medicinal Stems	0	10,000 Stems	14,000 Stems	18,000+ Stems
Sapling Survival Rate	0%	> 85%	> 90%	> 95% Established
Certified Junior Fellows	0	1,000 Students	2,500 Students	5,000+ Youth Stewards
Maiba/Maibi Masterclasses	0	50 Classes	150 Classes	250 Cumulative Classes
Digital QR Field Scans	0	5,000 Scans	25,000 Scans	60,000+ Scans
Annual Rainwater Recharged	0 Liters	750,000 Liters	850,000 Liters	1,000,000+ Liters

Formal Institutional Declaration & Endorsement

One Vision Manipur commits to implementing School Aushadhi Lab in full accordance with the guidelines of the National Medicinal Plants Board, Ministry of AYUSH. The organization guarantees complete financial transparency, timely submission of utilization certificates and audited statements, and permanent institutional care for all 50 school herbal sanctuaries.

PRESIDENT	GENERAL SECRETARY
Oinam Thoiba Singh President, One Vision Manipur	Thaibal Singh General Secretary, One Vision Manipur
PROJECT INVESTIGATOR	PROJECT COORDINATOR
Sapam Chital Singh B.Sc., D.M. College of Science	Lamnganbi Ngangom G.P. Women's College, Imphal

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